

Context Directions v. Auto Merchants, Inc.

Hierarchical sensor groups and adaptive classifiers for detecting a mobile device's context, alleged against Audi and Toyota vehicle systems.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-21	E.D. Tex.	2:26-cv-00729	Patent Infringement

Prepared 2026-08-25. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

Public filing, likely recipient, and technical frame

CAPTION	Context Directions LLC v. Auto Merchants, Inc.
FORUM / DOCKET	E.D. Tex. / 2:26-cv-00729
FILED / TYPE	2026-08-21 / patent infringement
PUBLIC DOCKET	Open docket

VERIFIED PUBLIC RECORD

Context Directions LLC v. Auto Merchants, Inc., No. 2:26-cv-00729, filed 2026-08-21 in E.D. Tex..

LIKELY RESEARCH PURCHASER

Steven G. Kalberg - Direction IP Law; named matter counsel. The docket identifies Kalberg as complaint filer and plaintiff counsel. Selecting the opening filer as likely purchaser is an analytical inference.

skalberg@directionip.com - publicly listed on the [official firm biography](#); not inferred.

TECHNICAL FRAME

Hierarchical sensor groups and adaptive classifiers for detecting a mobile device's context, alleged against Audi and Toyota vehicle systems.

Secondary-source limitation: This is a preliminary technical frame based on public sources. Filed pleadings, claim charts, discovery, source code, product versions, and admissible evidence must control. [Open public synopsis](#)

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 Whether accused Audi and Toyota systems organize heterogeneous sensors into the claimed hierarchy of sensor groups.

Whether accused Audi and Toyota systems organize heterogeneous sensors into the claimed hierarchy of sensor groups.

Potential evidence: Pleadings and claim charts; source code or engineering records; product and version documentation; standards and prior art; reproducible testing; and fact or technical testimony as applicable.

02 How context labels are produced, updated and propagated by adaptive classifiers, and where that processing occurs.

How context labels are produced, updated and propagated by adaptive classifiers, and where that processing occurs.

Potential evidence: Pleadings and claim charts; source code or engineering records; product and version documentation; standards and prior art; reproducible testing; and fact or technical testimony as applicable.

03 Whether power savings arise from selective sensor activation as claimed or from conventional vehicle power management.

Whether power savings arise from selective sensor activation as claimed or from conventional vehicle power management.

Potential evidence: Pleadings and claim charts; source code or engineering records; product and version documentation; standards and prior art; reproducible testing; and fact or technical testimony as applicable.

04 What source code, ECU logs, sensor traces and model artifacts are needed to distinguish claimed context detection from ordinary rules.

What source code, ECU logs, sensor traces and model artifacts are needed to distinguish claimed context detection from ordinary rules.

Potential evidence: Pleadings and claim charts; source code or engineering records; product and version documentation; standards and prior art; reproducible testing; and fact or technical testimony as applicable.

Questions likely to require expert input

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 Tanzeem Choudhury

Roger and Joelle Burnell Chair in Integrated Health and Technology and Professor, Cornell University

TECHNICAL FIT

Machine learning, mobile systems and activity recognition directly fit hierarchical sensor grouping and adaptive context classification. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Directs the People-Aware Computing group; published on mobile sensing, activity recognition and behavioral modeling.

PUBLIC LITIGATION EXPERIENCE

The reviewed Cornell profile and public CV do not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Most deployments concern health and human behavior rather than production automotive context engines or vehicle buses.

[Source 1](#) | [Source 2](#)

[Draft email](#) | tanzeem.choudhury@cornell.edu | [public email source](#)

02 Andrew T. Campbell

Albert Bradley 1915 Third Century Professor of Computer Science, Dartmouth College

TECHNICAL FIT

Longstanding mobile-sensing and AI research fits sensor fusion, learned context states, power-aware collection and real-world validation. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Leads mobile phone and wearable sensing studies and publishes on multimodal sensing and longitudinal behavior inference.

PUBLIC LITIGATION EXPERIENCE

The reviewed Dartmouth pages do not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Health-sensing orientation and use of a Gmail-address public contact may invite questions about automotive-specific implementation depth.

[Source 1](#) | [Source 2](#)

[Draft email](#) | andrewtcampbell@gmail.com | [public email source](#)

Candidates 3-4 for further diligence

03 Shwetak Patel

Washington Research Foundation Entrepreneurship Endowed Professor, University of Washington

TECHNICAL FIT

Ubiquitous sensing, embedded systems and activity recognition fit low-power sensor selection, context inference and deployment tradeoffs. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

ACM Prize recipient for practical sensing systems; founded companies commercializing sensing work.

PUBLIC LITIGATION EXPERIENCE

The reviewed UW profile and CV do not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Broad sensing expertise, but not specifically an automotive embedded-classifier or CAN-bus specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | shwetak@cs.washington.edu | [public email source](#)

04 David Kotz

Pat and John Rosenwald Professor of Computer Science, Dartmouth College

TECHNICAL FIT

Pervasive computing, wireless networks and smart-environment privacy fit sensor provenance, context inference and system architecture. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

More than 280 papers and 12 patents; research in pervasive computing, smart homes and wireless systems.

PUBLIC LITIGATION EXPERIENCE

His public page notes consulting through Harbor Experts, but the reviewed materials do not identify a specific testimony or litigation engagement.

POINTS FOR FURTHER DILIGENCE

Stronger on pervasive-system architecture and privacy than on adaptive-classifier mathematics or automotive production systems.

[Source 1](#) | [Source 2](#)

[Draft email](#) | David.F.Kotz@dartmouth.edu | [public email source](#)

Candidates 5-6 for further diligence

05 Anind Dey

Dean and Professor, Information School, University of Washington; adjunct faculty, Allen School

TECHNICAL FIT

Pioneer in context-aware and ubiquitous computing; fit for defining context, sensor hierarchies, adaptive models and user-facing behavior. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Research spans context-aware computing, active learning, behavior modeling and sensor-rich environments.

PUBLIC LITIGATION EXPERIENCE

The reviewed UW profiles do not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

HCI and ubiquitous-computing perspective may need supplementation for automotive hardware, firmware and claim-specific classifier implementation.

[Source 1](#) | [Source 2](#)

[Draft email](#) | anind@uw.edu | [public email source](#)

06 Deborah Estrin

Robert V. Tishman Professor of Computer Science and Associate Dean for Impact, Cornell Tech

TECHNICAL FIT

Embedded networked sensing and mobile-data analysis fit sensor architecture, real-time context data and power/collection tradeoffs. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Founding director of the NSF Center for Embedded Networked Sensing; pioneered mobile and wireless systems for real-time physical-world data.

PUBLIC LITIGATION EXPERIENCE

The reviewed Cornell profile does not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Current work emphasizes digital health; less direct experience with accused automotive subsystems and vehicle-specific sensors.

[Source 1](#)

[Draft email](#) | destrin@cornell.edu | [public email source](#)

Candidates 7-8 for further diligence

07 Mahadev Satyanarayanan

Jaime Carbonell University Professor of Computer Science, Carnegie Mellon University

TECHNICAL FIT

Mobile/pervasive computing and edge systems fit where sensor data is processed, latency/power allocation and adaptive device-cloud partitioning. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Pioneered cloudlets and edge computing; extensive work in mobile, pervasive and IoT systems.

PUBLIC LITIGATION EXPERIENCE

The reviewed CMU profile does not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

System architecture is stronger than statistical sensor-classifier design or automotive subsystem reverse engineering.

[Source 1](#) | [Source 2](#)

[Draft email](#) | satya@cmu.edu | [public email source](#)

08 Julie Kientz

Professor and Chair, Human Centered Design & Engineering, University of Washington

TECHNICAL FIT

Mobile, sensor and social applications fit evaluation of context recognition, user burden and sensor-driven adaptive behavior. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Designed and evaluated mobile and sensor applications across health, education and family domains.

PUBLIC LITIGATION EXPERIENCE

The reviewed UW profiles do not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Human-centered evaluation strength is less direct for low-level classifier architecture, automotive firmware and infringement mapping.

[Source 1](#) | [Source 2](#)

[Draft email](#) | jkientz@uw.edu | [public email source](#)

Candidates 9-10 for further diligence

09 Alex Pentland

Toshiba Professor of Media Arts and Sciences, MIT Media Lab

TECHNICAL FIT

Wearable computing, smart environments and human-dynamics sensing fit multimodal context recognition and adaptive inference. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Pioneer in wearable computers, smart environments and sensing/modeling human behavior.

PUBLIC LITIGATION EXPERIENCE

The reviewed MIT public technical materials do not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Human-dynamics emphasis is less direct for automotive embedded implementation, ECU interfaces or accused-source-code analysis.

[Source 1](#) | [Source 2](#)

[Draft email](#) | pentland@mit.edu | [public email source](#)

10 Cecilia Mascolo

Professor of Mobile Systems, University of Cambridge

TECHNICAL FIT

Mobile systems, sensor platforms and machine learning fit context sensing, on-device classification and power-aware mobile inference. Matter-specific fit: Audi and Toyota vehicle context detection, hierarchical sensor groups and adaptive classifiers.

RELEVANT WORK

Research and publications cover mobile sensing systems, phone/wearable sensing and real-world deployments.

PUBLIC LITIGATION EXPERIENCE

The reviewed Cambridge profile does not identify expert-testimony work.

POINTS FOR FURTHER DILIGENCE

Non-U.S. academic and health/mobile focus may require added automotive-domain support and travel/logistics diligence.

[Source 1](#) | [Source 2](#)

[Draft email](#) | cm542@cam.ac.uk | [public email source](#)

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.
- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.
- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.
- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

[Schedule a 15-minute conversation](#)

CORE SOURCES

Public docket: <https://dockets.justia.com/docket/texas/txedce/2%3A2026cv00729/248608>

Official attorney biography and email: <https://ai-lab.exparte.com/attorney/6517108/steven-g-kalberg>

Public technical context source: <https://ai-lab.exparte.com/case/dct/txed/2%3A26-cv-00729/doc/analysis/1>

LIMITATIONS

This is a preliminary technical frame based on public sources. Filed pleadings, claim charts, discovery, source code, product versions, and admissible evidence must control.

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